recently we had never performed the obvious calibration test: to fly a modern interplanetary spacecraft by the Earth and see whether we could detect ourselves. This all changed on December 8, 1990.

Galileo is a NASA spacecraft designed to explore the giant planet Jupiter, its moons, and its rings. It's named after the heroic Italian scientist who played so central a role in toppling the geocentric pretension. It is he who first saw Jupiter as a world, and who discovered its four big moons. To get to Jupiter, the spacecraft had to fly close by Venus (once) and the Earth (twice) and be accelerated by the gravities of these planets—otherwise there wasn't enough oomph to get it where it was going. This necessity of trajectory design permitted us, for the first time, to look systematically at the Earth from an alien perspective.

Galileo passed only 960 kilometers (about 600 miles) above the Earth's surface. With some exceptions—including pictures showing features finer than 1 kilometer across, and the images of the Earth at night—much of the spacecraft data described in this chapter were actually obtained by *Galileo*. With *Galileo* we were able to deduce an oxygen atmosphere, water, clouds, oceans, polar ice, life, and intelligence. The use of instruments and protocols developed to explore the planets to monitor the environmental health of our own —something NASA is now doing in earnest—was described by the astronaut Sally Ride as "Mission to Planet Earth."

Other members of the NASA scientific team who worked with me on *Galileo*'s detection of life on Earth were Drs. W. Reid Thompson, Cornell University; Robert Carlson, JPL; Donald Gurnett, University of Iowa; and Charles Hord, University of Colorado.

Our success in detecting life on Earth with *Galileo*, without making any assumptions beforehand about what kind of life it must be, increases our confidence that when we fail to find life on other planets, that negative result is meaningful. Is this judgment anthropocentric, geocentric, provincial? I don't think so. We're not looking only for our kind of biology. Any widespread photosynthetic pigment, any gas grossly out of equilibrium with the rest of the atmosphere, any rendering of the surface into highly geometrized patterns, any steady constellation of lights on the night hemisphere,